

STAT 7650. Homework 5 (Due: Tuesday, April 14, 2026)

1. (10 points) Let $Y = X/U$, where $X \sim N(0, 1)$ and $U \sim \text{unif}(0, 1)$ are independent. The distribution of Y is called the slash distribution with density

$$f(y) = \begin{cases} \frac{1 - \exp(-y^2/2)}{y^2 \sqrt{2\pi}}, & y \neq 0 \\ \frac{1}{2\sqrt{2\pi}}, & y = 0 \end{cases}$$

- (a) Use the slash distribution as a SIR (sampling important resampling) envelope to generate standard normal random variables. Set $m = 100,000$ and $n = 5,000$. Draw a histogram of the sample along with the normal density curve.
- (b) Use the standard normal distribution as a SIR (sampling important resampling) envelope to generate slash random variables. Set $m = 100,000$ and $n = 5,000$. Draw a histogram of the sample along with the slash density curve.
2. (10 points) Use two different methods to compute the Monte Carlo estimate of the probability $P(Z > 4.5)$, where $Z \sim N(0, 1)$, that is,

$$P(Z > 4.5) = \int_{4.5}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

- (a) Generate $n = 1,000$ samples from the $N(0, 1)$ distribution. Compute the Monte Carlo estimate of the probability and its standard error.
- (b) Generate $n = 1,000$ samples from the shifted exponential distribution with density $f(y) \propto e^{-(y-4.5)}$ for $y > 4.5$. Compute the Monte Carlo estimate of the probability and its standard error. (Hint: use importance sampling)
3. (10 points) Let X be a random variable with density $f(x) \propto \exp(-|x|^3/3)$. Use the following two different methods to compute the Monte Carlo estimate of $\sigma^2 = E(X^2)$.
- (a) Use importance sampling with $N(0, 1)$ distribution and standardized weights. Compute the estimate and its standard error using $n = 1,000$ samples.
- (b) Generate a sample of $n = 1,000$ for X directly using rejection sampling with $N(0, 1)$ proposal distribution. Compute the estimate and its standard error.
4. (20 points) Consider testing the hypothesis $H_0 : \lambda = 2$ versus $H_a : \lambda > 2$ using 25 observations from a $\text{Poisson}(\lambda)$ distribution. We may reject H_0 at the significance level $\alpha = 0.05$ when $Z = (\bar{X} - 2)/\sqrt{2/25} \geq 1.645$, using normal approximation. The actual size of this test (type I error rate) is

$$\alpha = P_{\lambda=2}(Z \geq 1.645).$$

Compute the actual size α and its standard error using different Monte Carlo methods with $n = 1,000$.

- (a) Use the standard Monte Carlo method, that is, sample from Poisson distribution directly.
 - (b) Use the antithetic sampling. Notice that `qpois(u, lambda)` returns a Poisson random variable if $U \sim \text{uniform}(0, 1)$.
 - (c) Use the importance sampling with unstandardized and standardized weights, respectively. Sample from a Poisson distribution with mean $\lambda = 2.4653$, which is the mean equal to the H_0 rejection threshold.
 - (d) Use the importance sampling with a control variate, $\hat{\mu}_{\text{ISCV}} = \hat{\mu}_{\text{IS}}^* + c(\bar{w}^* - 1)$, where $\hat{\mu}_{\text{IS}}^*$ is the Monte Carlo estimate using importance sampling with unstandardized importance weights, $\bar{w}^*$ is the average of the unstandardized importance weights, and c is the constant that minimizes $\text{var}(\hat{\mu}_{\text{ISCV}})$.
5. (10 points) Let $X \sim \text{log-normal}(0, 1)$ and $\log Y = 9 + 3 \log X + \varepsilon$, where $\varepsilon \sim N(0, 1)$. Estimate $E(Y/X)$ using two different methods with $n = 1,000$ samples. Also compute the standard errors of the estimates.
- (a) Use the standard Monte Carlo estimator.
 - (b) Use the Rao-Blackwellized estimator.
6. (10 points) Let $Y_1, \dots, Y_n$ be a sample from a mixture of two normal distributions

$$Y \sim \delta N(7, 0.5^2) + (1 - \delta)N(10, 0.5^2),$$

where $\delta \in (0, 1)$ is an unknown parameter. The prior distribution of δ is $\text{uniform}(0, 1)$. In the following questions, we will use different methods to draw from the posterior distribution of δ . You may generate a sample of size $n = 100$ using $\delta = 0.7$ as the observed data.

- (a) Use independence chains, where the proposal distribution is either $\text{Beta}(1, 1)$ or $\text{Beta}(2, 10)$. Draw the trace plots, autocorrelation plots, and histograms of 10,000 posterior samples. Compare the plots with all samples and those after discarding the initial half posterior samples as burn-in.
 - (b) Use a random walk applied to the logit transformation of δ . The proposal distribution is $\text{uniform}(-a, a)$ with $a = 1$ or $a = -0.01$. Draw plots to check convergence similarly as in the previous question.
7. (10 points) Use the capture-recapture method to estimate the population N of pups in a fur seal colony. The individuals captured during each census are released with a marker indicating their capture. The following table lists the total number of captured pups and the number of newly captured pups for $I = 7$ census in one sampling period.

Census i	1	2	3	4	5	6	7
Number captured c_i	30	22	29	26	31	32	35
Number newly caught m_i	30	8	17	7	9	8	5

The likelihood function is

$$L(N, \alpha \mid c, r) \propto \frac{N!}{(N-r)!} \prod_{i=1}^I \alpha_i^{c_i} (1 - \alpha_i)^{N-c_i},$$

where N is the unknown population size, I is the total number of census attempts, $c = (c_1, \dots, c_I)$ is the number of captures, $\alpha = (\alpha_1, \dots, \alpha_I)$ is the unknown capture probabilities, and r is the total number of distinct pups captured during the study. The prior distribution of N is uniform and the prior distribution of α_i is $\text{Beta}(\theta_1, \theta_2)$ for $i = 1, \dots, I$.

- (a) Set $\theta_1 = \theta_2 = 0.5$. Draw 100,000 samples of N and α using Gibbs sampling and discard the first 50,000 samples for burn-in. Check convergence using the trace plots and autocorrelation plots. Compute the effective sample size, 95% HPD intervals for each parameters.
- (b) We further assume a prior for θ_1 and θ_2 ,

$$f(\theta_1, \theta_2) \propto \exp\{(\theta_1 + \theta_2)/1000\}.$$

The posterior distribution of θ_1 and θ_2 is not standard. Hence, we sample (θ_1, θ_2) using a random walk with $N(0, 0.085^2)$ after log-transformation. The whole sampling algorithm is essentially a Metropolis-within-Gibbs. Draw a sample of 100,000 and discard the first 50,000 as burn-in. Then draw plots and compute summaries as in the previous question.

8. (20 points) The datafile `pigment.txt` contains the moisture contents from 15 batches of pigment paste. Each batch of the pigment was sampled twice independently. The unit is one tenth of 1%.

<https://github.com/pzengauburn/STAT-7650-CompStat>

Consider a hierarchical nested model

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{ijk}, \quad i = 1, \dots, I, \quad j = 1, \dots, J_i, \quad k = 1, \dots, K.$$

After averaging over k for each i and j , we can rewrite the model as

$$Y_{ij} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{ij}, \quad i = 1, \dots, I, \quad j = 1, \dots, J_i,$$

where $Y_{ij} = \sum_{k=1}^K Y_{ijk}/K$. Assume that $\alpha_i \sim N(0, \sigma_\alpha^2)$, $\beta_{j(i)} \sim N(0, \sigma_\beta^2)$, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$, where each set of parameters is independent a priori. Assume that σ_α^2 , σ_β^2 , and σ_ε^2 are known. We further assume an improper flat prior for μ , so $f(\mu) \propto 1$. Introduce notations,

$$n = \sum_i J_i, \quad \bar{y}_{..} = \sum_i \sum_j y_{ij}/n, \quad \bar{y}_{i.} = \sum_j y_{ij}/J_i.$$

- (a) Derive the Gibbs sampling by working out the conditional posterior distributions for each parameter of μ , α_i , $\beta_{j(i)}$.
- (b) Let $\sigma_\alpha^2 = 86$, $\sigma_\beta^2 = 58$, and $\sigma_\varepsilon^2 = 1$. Implement the Gibbs sampler in block. For example, $\alpha = (\alpha_1, \dots, \alpha_{15})$ is one block where all parameters can be updated simultaneously because their conditional distributions are independent. After deleting the burn-in iterations, draw plots to check convergence and compute effective sample sizes for μ , α_1 , and $\beta_{1(1)}$.
- (c) The convergence rate for the Gibbs sampler can be improved by reparameterization via hierarchical centering. Let $\eta_{ij} = \mu + \alpha_i + \beta_{j(i)}$ and $\gamma_i = \mu + \alpha_i$. Then

$$\eta_{ij} \mid \gamma_i \sim N(\gamma_i, \sigma_\beta^2), \quad \gamma_i \mid \mu \sim N(\mu, \sigma_\alpha^2).$$

Derive the Gibbs sampling for each parameter of μ , γ_i , and η_{ij} .

- (d) Implement the Gibbs sampler. After deleting the burn-in iterations, draw plots to check convergence and compute effective sample sizes for μ , γ_1 , and η_{11} .